

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes

Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Naveen ms (192371002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

naveen

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand
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1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Smart Home Automation System Using Intelligent Agents:

A smart home automation system controls **lighting, temperature, and security** by using information from sensors such as motion detectors, temperature sensors, and clocks. It makes decisions automatically to improve **comfort, energy efficiency, and home security**. Different types of intelligent agents can be used in such a system.

1. Simple Reflex Agent

A **Simple Reflex Agent** makes decisions based only on the current input. It follows predefined **if-then rules** without considering previous events.

Example:

- If motion is detected → Turn on the lights.
- If temperature is above 30°C → Turn on the air conditioner.
- If no motion is detected for 10 minutes → Turn off the lights.

Advantages:

- Fast response.
- Easy to design.
- Suitable for simple tasks.

Disadvantages:

- Cannot remember past events.
- Cannot adapt to user preferences.
- May waste energy if rules are not sufficient.

2. Model-Based Agent

A **Model-Based Agent** keeps an internal model of the environment. It remembers previous states and uses that information to make better decisions.

Example:

- If the living room was occupied recently, keep the lights on for a few minutes.
- Monitor the temperature trend before switching the AC on or off.
- Detect unusual movement when the house is normally empty.

Advantages:

- Considers past information.
- Makes more accurate decisions.
- Better for changing environments.

Disadvantages:

- More complex than a simple reflex agent.
- Requires memory and additional processing.

3. Goal-Based Agent

A **Goal-Based Agent** chooses actions that help achieve specific goals.

Goals in a Smart Home:

- Maintain a comfortable room temperature.
- Reduce electricity consumption.
- Ensure home security.
- Keep all rooms properly lit when occupied.

Example:

If the goal is to save energy, the agent turns off unused appliances while maintaining user comfort.

Advantages:

- Makes intelligent decisions based on goals.
- Can plan different actions to achieve desired results.

Disadvantages:

- Planning takes more time.
- More computational resources are required.

4. Utility-Based Agent

A **Utility-Based Agent** selects the action that provides the highest overall benefit by comparing different choices using a utility value.

Example:

The system considers:

- User comfort
- Electricity cost
- Energy consumption
- Security level

If electricity prices are high, it slightly increases the AC temperature while still keeping the room comfortable. It balances comfort with energy savings.

Advantages:

- Provides the best overall decision.
- Balances multiple objectives.
- Learns and adapts to user preferences over time.

Disadvantages:

- More difficult to design.
- Requires utility calculations and learning mechanisms.

A **hybrid approach** combining **Model-Based**, **Goal-Based**, and **Utility-Based Agents** is the most effective solution.

- The **Model-Based Agent** remembers previous conditions and user behavior.
- The **Goal-Based Agent** ensures objectives such as comfort, security, and energy efficiency are achieved.
- The **Utility-Based Agent** selects the best action by balancing comfort, energy usage, electricity cost, and security.

The **Simple Reflex Agent** can still be used for quick actions such as switching on lights immediately when motion is detected.

Conclusion

A **hybrid intelligent agent** is the best choice for a smart home automation system because it combines fast responses, environmental awareness, goal achievement, and decision optimization. It learns user preferences over time and balances **comfort, energy efficiency, and security**, making the home more intelligent and efficient.

2 robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects

decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Robot Navigation in an Unknown Maze Using Uninformed Search

A robot placed in an unknown maze must explore the environment and find a path to the exit without knowing the maze layout in advance. To solve this problem, the robot uses **uninformed (blind) search algorithms**, which do not have any knowledge about the location of the goal. Two commonly used uninformed search strategies are **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**.

1. Uniform Cost Search (UCS)

Uniform Cost Search (UCS) expands the path with the **lowest cumulative cost** first. It uses a **priority queue** to select the next node based on the total path cost.

Working

- Start from the initial position.
- Expand the node with the lowest path cost.
- Continue until the exit is reached.
- Always finds the least-cost path if all costs are positive.

Example

If different paths in the maze have different movement costs (such as stairs, rough terrain, or obstacles), UCS selects the path with the minimum total cost instead of the shortest number of steps.

Advantages

- Finds the **optimal (least-cost)** solution.
- Complete if path costs are positive.
- Suitable when movement costs are different.

Disadvantages

- Requires large memory to store all generated nodes.
- Can be slow in large mazes.
- Higher computational cost.

Complexity

- **Time Complexity:** $O(b^{1+\lceil C^*/\epsilon \rceil})$
- **Space Complexity:** $O(b^{1+\lceil C^*/\epsilon \rceil})$

Where:

- **b** = branching factor
- **C*** = optimal solution cost
- **ε** = minimum step cost

2. Iterative Deepening Search (IDS)

Iterative Deepening Search (IDS) combines the advantages of **Depth-First Search (DFS)** and **Breadth-First Search (BFS)**. It repeatedly performs depth-limited searches with increasing depth until the goal is found.

Working

- Search at depth 0.
- If the goal is not found, increase the depth limit to 1.
- Continue increasing the depth until the exit is found.

Example

The robot first searches nearby locations. If the exit is not found, it explores deeper levels gradually.

Advantages

- Requires very little memory.
- Complete for finite search spaces.
- Finds the shallowest solution.

Disadvantages

- Repeats exploration of upper levels.
- Not suitable when path costs differ.
- Can take longer due to repeated searches.
- **Complexity**

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(bd)$

- Where:

- **b** = branching factor
- **d** = depth of the solution

Relation to Intelligent Agents

Different intelligent agents influence how the robot makes decisions while exploring the maze.

Simple Reflex Agent

- Responds only to the current surroundings.
- Cannot remember previous paths.
- May repeatedly visit the same locations.
- Not suitable for unknown mazes.

Model-Based Agent

- Maintains an internal map of explored paths.
- Avoids revisiting dead ends.
- Makes better navigation decisions.

Goal-Based Agent

- Focuses on reaching the maze exit.
- Uses search algorithms such as UCS or IDS to achieve the goal.
- Selects actions that move closer to the destination.

Effect of Agent Design on Decision-Making

The design of the intelligent agent directly affects the robot's performance.

- **Simple Reflex Agent** reacts quickly but cannot learn or remember previous moves.
- **Model-Based Agent** stores information about explored areas and reduces unnecessary exploration.
- **Goal-Based Agent** plans actions to reach the exit efficiently.
- **Utility-Based Agent** evaluates different paths and selects the one with the highest overall benefit, considering time, cost, and energy.

Better agent designs lead to improved navigation, fewer repeated movements, and faster completion of the task.

Conclusion

For navigating an unknown maze, **Uniform Cost Search (UCS)** is suitable when movement costs vary because it always finds the optimal path. **Iterative Deepening Search (IDS)** is preferable when memory is limited and all paths have equal cost. A **Goal-Based Agent**, supported by a **Model-Based Agent** for remembering explored paths, provides the most effective decision-making. This combination enables the robot to navigate efficiently, avoid repeated exploration, and successfully reach the exit with optimal performance.